Claudia Martens [DKRZ]
Alexander Wolodkin [Senckenberg]

Combining Machine Learning tools and Terminology Services to support FAIR Data Management in ESS

Lightning Talk at IDCC26
Feb 17th, 2026

BITS
BluePrints for the Integration of Terminology Services in Earth System Sciences

CC BY

Data in Earth System Science

divergent
complex
huge!

Data in Earth System Science

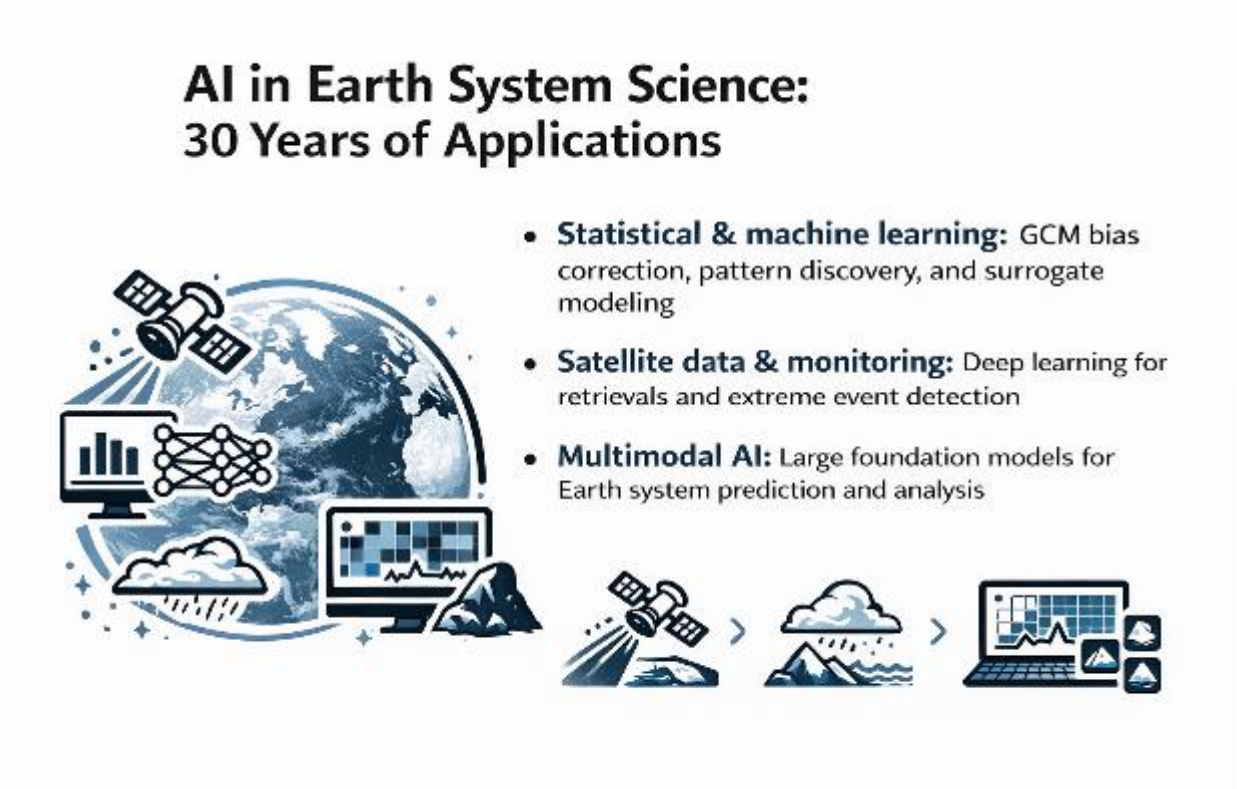
- **Different Data Types**
  model output, observations, simulations, measurements

- **Different Formats**
  NetCDF (HDF5), GeoTIFF, GRIB, GeoJSON, ASCII/CSV or frameworks like GIS, tabular data, text, images

- **Different Sizes / Granularity Level**
  from PB in radio astronomy and climate model output to MB in samples in Biology

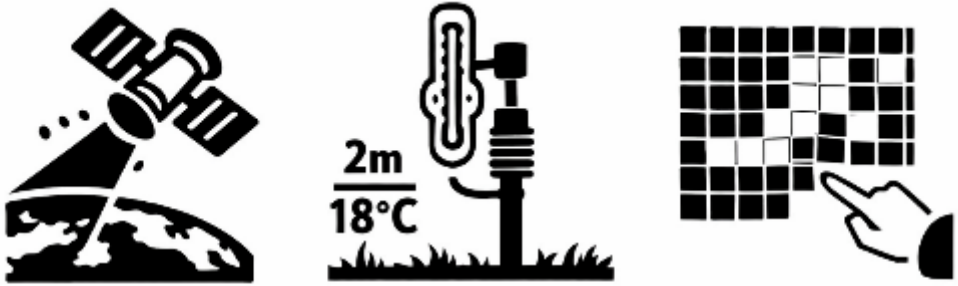
AI in Earth System Science

- AI is the new ML
- From statistical methods to Neuronal Networks
- What is new: it scales!



One Term – Three Realities

Surface Temperature



Satellite Radiance

In-situ Air Temperature at 2 Meters

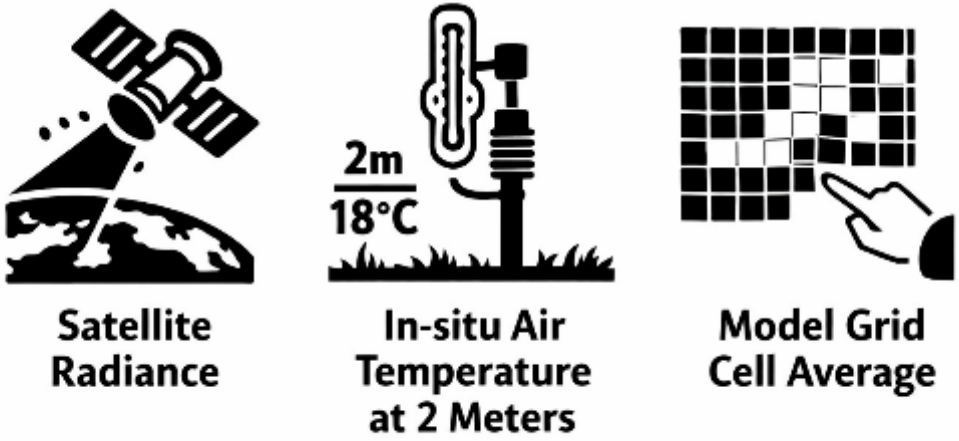
Model Grid Cell Average

One Term – Three Realities

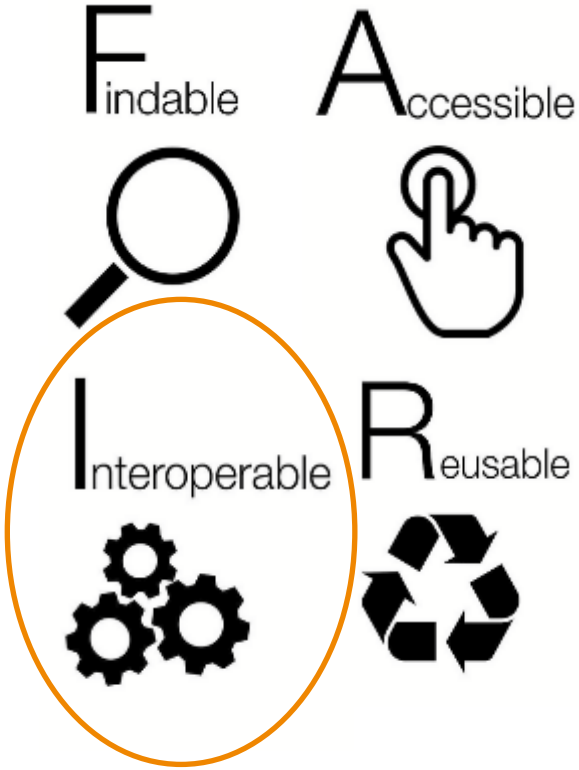
- **Satellite radiance**
  measures outgoing thermal energy
- **In-situ air temperature at 2m**
  standard meteorological observation
- **Model grid cell average**
  numerical approximation over a discrete area

→ Semantic Heterogenity

Surface Temperature



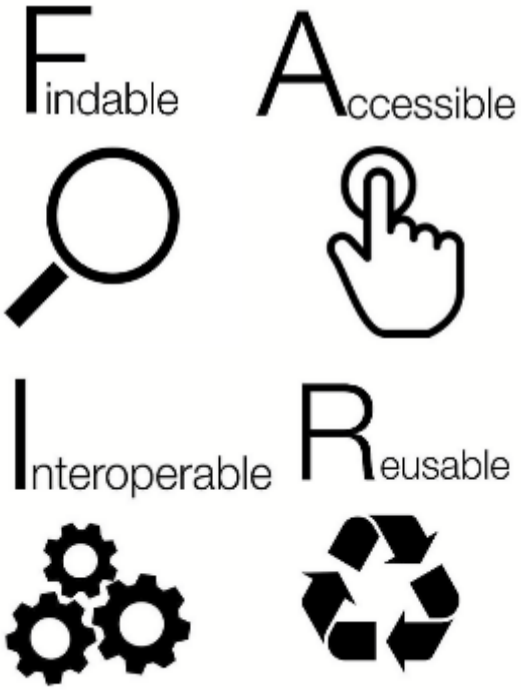
The FAIR - Bottleneck



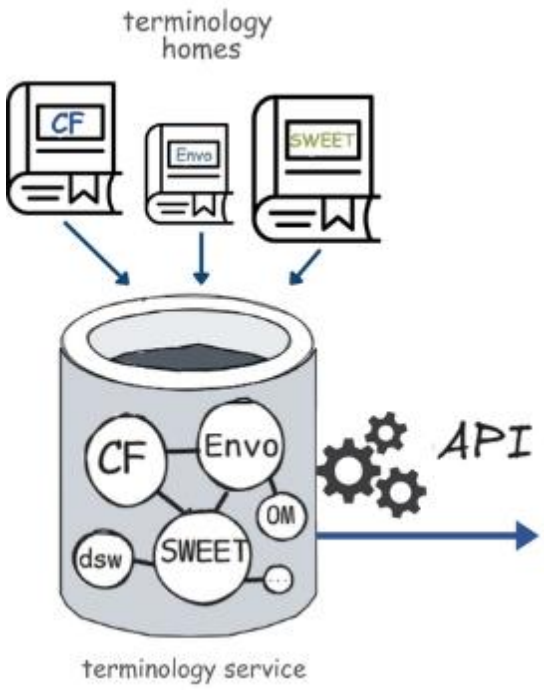
The FAIR - Bottleneck

- **Shared Meaning Across Systems**
  = giving context and consistency using standardized vocabularies, ontologies, and metadata schemas
- **Use of Controlled Vocabularies and Ontologies**
  = relies on community-agreed terms and structures to describe data, enabling accurate interpretation, integration, and automated reasoning across disciplines
- **Supports FAIR and Cross-Domain Research**
  = enhances data reusability and integration across domains, supporting the FAIR principles and enabling interdisciplinary environmental research

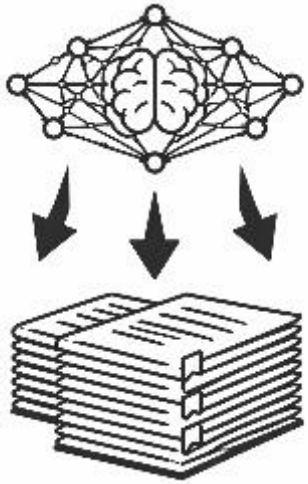
→ Semantic Interoperability



Terminology Services vs Machine Learning



- Automated annotation can process millions of records
- Pattern extraction is fast and consistent
- Risk: scaling ambiguity instead of clarity



- Controlled vocabularies define and disambiguate terms
- Persistent identifiers ensure reference stability
- API access allows machine workflows
- Versioning tracks evolution of concepts
- Problem: they don´t scale

Bring it together!

Blueprint for the **I**ntegration of **T**erminology **S**ervices in ESS

Already existing and reliable service = ESS Collection of TIB Terminology Service

TIB TERMINOLOGY SERVICE

HOME COLLECTIONS ONTOLOGIES HELP API ABOUT

Search for ontology, term, properties and individuals | Search

Exact match | Obsolete terms | Include imported terms | Advanced search

ESS EARTH SYSTEM SCIENCES

Earth System Sciences

The ESS Terminology Service is a collection of Earth System Science terminologies and a single point of access to the latest terminology versions. The service provides researchers, repository staff, and everybody interested with the ability to describe their research results in a semantically correct, machine-understandable way. The ESS TS is being developed by the BITS project titled "BluePrints for the Integration of Terminology Services in Earth System Sciences" and is maintained by TIB as an extension of the TIB Central Terminology Service. It can be used either by humans through the website or by machines via the TS API.

Project Homepage: https://projects.tib.eu/bits

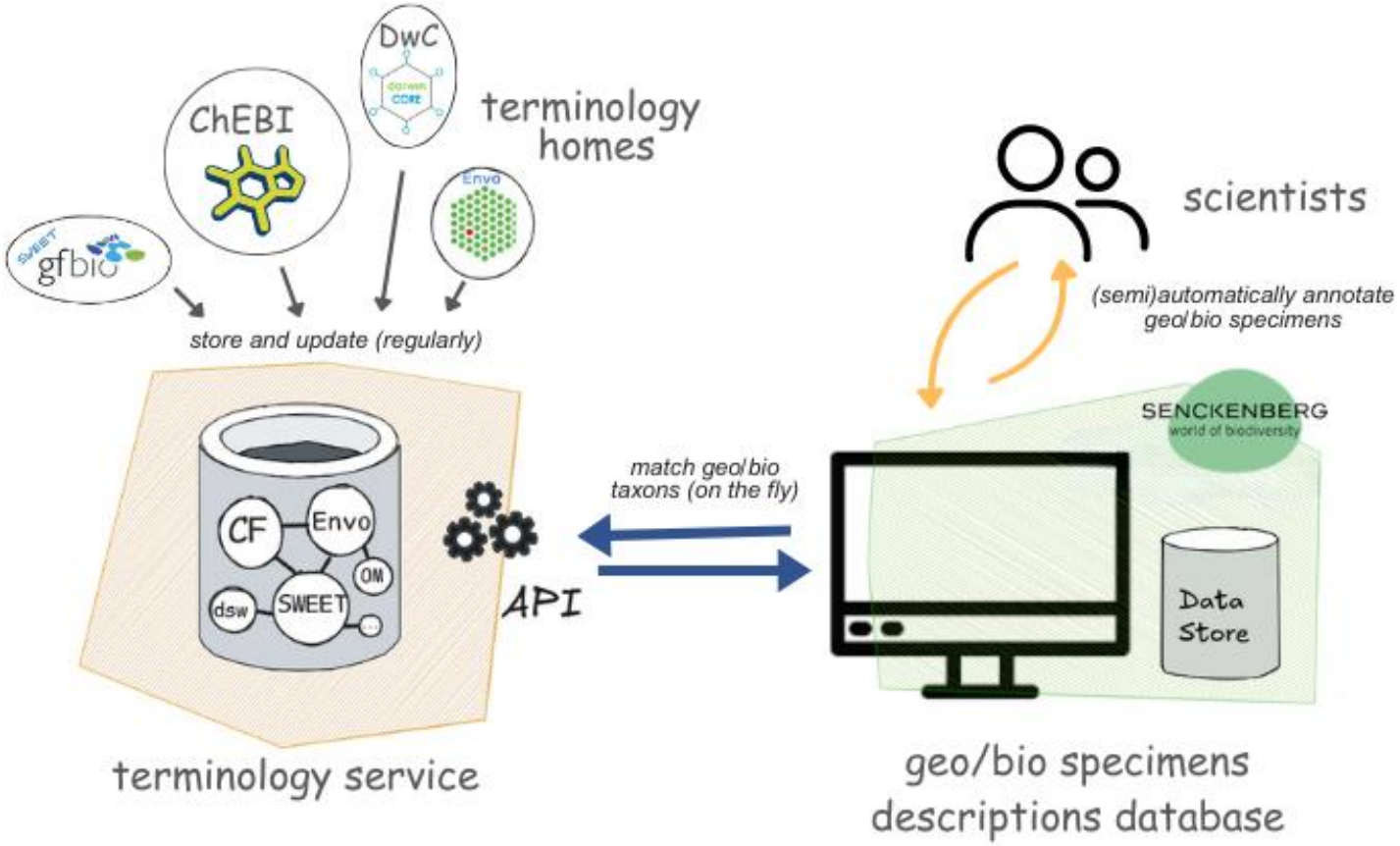
Ontologies: sio sbo uo dcterms abcd cf sosa dsw uat bk efo ets ecso time om to dfgfo envo gndo prov envo2021 e t stato dfgfo2024 envo2023 edam t4fs schema pato ssn oboe-standards gemet oboe bco geo oboe-core dpv foaf meth dcat3 po n4e wlmo oboe-characteristics duo sdn sweet chebi Suggest an ontology for this collection

Example

SENCKENBERG
world of biodiversity

- Natural history museum and research center
- currently 1.529.933 digital objects in 124 collections
- Automated annotation with LLMs (finding keywords in texts) and standardisation using TIB TS terminologies.

Example


DwC
ChEBI
terminology homes
Envo
gfbio
store and update (regularly)
scientists
(semi)automatically annotate geo/bio specimens
SENCKENBERG
world of biodiversity
match geo/bio taxons (on the fly)
CF
Envo
OM
dsw
SWEET
API
Data Store
terminology service
geo/bio specimens descriptions database

Conclusion and Outlook

- Heterogeneity in ESS data is unavoidable
- Shared semantics increase interoperability
- AI tools need semantic grounding for FAIR data

FAIR is about shared meaning, not just access → Integration of AI + Terminology is essential

Reducing Ambiguity, Not Complexity

BITS 2.0

- Follow-up proposal
- Establish a Semantic Federation for Earth System Sciences (ESS)
  to interconnect heterogeneous data infrastructures through shared, machine-interpretable semantics
- Develop AI-based Tools
  for automated semantic annotation and metadata enrichment across diverse ESS repositories.
- Enhance Semantic interoperability, Data Quality, and AI-Readiness via structured, provenance-aware metadata
- Deliver modular, reusable “Blueprints 2.0”
  for scalable integration into existing large and small data holdings
- Implement a Sustainable Governance and Terminology Curation Framework

Conclusion and Outlook

BITS 2.0

THANK YOU!

https://projects.tib.eu/bits/home

TIB LEIBNIZ INFORMATION CENTRE FOR SCIENCE AND TECHNOLOGY UNIVERSITY LIBRARY

SENCKENBERG world of biodiversity

DKRZ DEUTSCHES KLIMARECHENZENTRUM

Questions?

funded by: DFG Deutsche Forschungsgemeinschaft

CC BY